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Member for adjusting force application in reciprocating assembly

Abstract

A reciprocating assembly includes a bore that contains hydraulic fluid. A piston is received within the bore and can move within the bore in a reciprocating motion between an extended position and a retracted position. Displacement of the piston within the bore causes displacement of hydraulic fluid when hydraulic fluid is in the bore. The piston includes an internal cavity extending at least partially along a length of the piston. The reciprocating assembly also includes a push pin that is received within the internal cavity. The bore, piston, and push pin are arranged so that movement of the push pin related to the bore causes translation of the piston within the bore to the extended position along a translation axis. The push pin can pivot during movement of the push pin between an in-line position and an angled position. The in-line position is substantially parallel to the translation axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Patent Application No. 63/427,579, filed Nov. 23, 2022, the entire contents of which is hereby incorporated by reference.

FIELD

(1) The present disclosure relates to piston assembly. More particularly, the present disclosure relates to a push pin for directing a force applied to a piston used in a reciprocating assembly.

BACKGROUND

(2) Reciprocating assemblies, like gas engines, hydraulic pumps, and pneumatic cylinders among other things, utilize one or more pistons to move a working fluid (e.g., gasoline, oil, air, etc.). The piston is received within a cylinder and moves back and forth in a reciprocating motion. In a retracted position, the working fluid may fill the cylinder. In an extended position, the piston may push the working fluid out of the cylinder.

(3) To move to the extended position, a force must be applied to the piston, which must overcome a force within the cylinder. In some examples, like an internal combustion engine, the force must be enough to compress the working fluid (e.g., gasoline). In other examples, the piston may need to

overcome a spring force or some other oppositely directed force.

(4) The reciprocating motion may be provided by a rotating cam and a wobble plate. The inclination of the rotating cam may cause the wobble plate to rock back and forth. This rocking motion creates the reciprocating motion to move the piston forward and backward within the cylinder.

(5) Because the cam, and therefore the wobble plate, is angled relative to the piston, the force provided by the wobble plate may include components in multiple directions. In other words, the force may not be directed entirely along the axis of the cylinder. This results in inefficiencies in the system because a greater resultant force is needed to supply the necessary component force directed along the cylinder. Additionally, force components in other directions (e.g., perpendicular to the axis of the cylinder) may drive the piston against the wall of the cylinder, particularly while the piston moves forward toward the end of the cylinder. This may lead to frictional wear between the cylinder and the piston, which can degrade either of the pieces and ultimately lead to a failure.

SUMMARY

(6) Various examples of the present disclosure can overcome the aforementioned and other disadvantages associated with known piston assemblies and offer new advantages as well.

(7) According to one aspect of various examples of the present disclosure there is provided a piston assembly that directs force substantially along an axis of the cylinder.

(8) According to another aspect of various examples of the present disclosure, there is provided a piston having a hollow center and a push pin received within the hollow center. The push pin adapted to apply a force within the hollow center in order to reduce a perpendicular component of force.

(9) According to another aspect of various examples of the present disclosure, there is provided a reciprocating assembly for driving a working portion of a tool, the reciprocating assembly comprising: a piston including an internal cavity extending at least partially along the length of the piston; and a push pin received within the internal cavity; wherein movement of a drive assembly is configured to supply a force for moving the piston, the force being applied to the push pin to translate the piston; and wherein the push pin is configured to pivot as the drive assembly moves between an in-line position and an angled position, the in-line position being substantially parallel to an axis along which the piston moves.

(10) According to another aspect of various examples of the present disclosure, there is provided a reciprocating assembly for driving a working portion of a tool, the reciprocating assembly comprising: a bore configured to contain hydraulic fluid; a piston received within the bore and configured to move within the bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the piston within the bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the bore, the piston including an internal cavity extending at least partially along a length of the piston; and a push pin received within the internal cavity of the piston; wherein the bore, the piston, and the push pin are configured such that a movement of the push pin relative to the bore causes translation of the piston within the bore to the extended position along a translation axis; and wherein the push pin is configured to pivot during movement of the push pin between an in-line position and an angled position, the in-line position being substantially parallel to the translation axis.

(11) In some forms: a) the cavity extends along a majority of the length of the piston; b) the push pin includes a cutout at each of two opposite ends; c) each cutout is configured to at least partially receive a ball bearing; and/or d) one of the ball bearings is received within the internal cavity and the other of the ball bearings is coupled to a drive assembly configured to supply a force which causes the movement of the push pin.

(12) In some forms: a) the push pin includes two opposite hemispherically shaped ends, wherein one of the hemispherically shaped ends is received within the internal cavity and the other of the hemispherically shaped ends is coupled to a drive assembly configured to supply a force which

causes the movement of the push pin; b) the push pin includes a hyperboloid shape; c) the internal cavity includes a terminal end and an open end, the open end being wider than the terminal end; d) the open end has a frustoconical shape; and/or e) a width of the open end is greater than a total distance of travel of the push pin.

(13) In some forms: a) an O-ring is connected to an outer surface of the push pin; b) the O-ring is compressible; c) a drive assembly can supply a force which causes the movement of the push pin; and/or d) the bore is configured to receive hydraulic fluid from a reservoir.

(14) In some forms: a) a drive assembly includes an electric motor, a gear assembly configured to be driven by the electric motor, an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly, and a wobble plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; b) the wobble plate is configured to not rotate with the inclined plate; c) an anti-rotational assembly is connected between the wobble plate and a housing containing the bore; d) the anti-rotational assembly can limit rotation of the wobble plate and permit axial movement of the wobble plate; e) the anti-rotational assembly includes at least one spring that extends axially between the wobble plate and the housing; f) the anti-rotational assembly includes at least one spring that extends axially between the wobble plate and the housing; g) the anti-rotational assembly includes a pin that extends radially between the wobble plate and the housing; h) the outer perimeter of the wobble plate includes a series of teeth and grooves; and/or i) the pin is configured to fit in a groove between adjacent teeth.

(15) A reciprocating assembly for driving a working portion of a tool, the reciprocating assembly including: a bore configured to contain hydraulic fluid; a piston received within the bore and configured to move within the bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the piston within the bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the bore, the piston including an internal cavity extending at least partially along a length of the piston; and a push pin having a first end received within the internal cavity of the piston, the first end having a first radius of curvature; wherein the push pin is configured to pivot about a center of the first radius of curvature between an in-line position and an angled position, the in-line position substantially parallel to a translation axis of the piston; and wherein the piston is configured to move from the retracted position toward the extended position when the push pin is in the in-line position.

(16) In some forms: a) the first end includes a first cutout having the first radius of curvature; b) the first cutout at least partially receiving a first ball bearing; c) the push pin is configured to rotate about the first ball bearing; d) a second end of the push pin opposite the first end includes a second cutout having a second radius of curvature; e) the second cutout at least partially receiving a second ball bearing; and/or f) the second ball bearing is coupled to a drive assembly can supply a force which causes the movement of the push pin.

(17) In some forms: a) the push pin includes a hyperboloid shape; b) the internal cavity includes a terminal end and an open end; c) the open end being wider than the terminal end; d) an O-ring is connected to an outer surface of the push pin; e) the O-ring is compressible; and/or f) wherein the first end is hemispherically shaped.

(18) In some forms: a) a drive assembly can supply a force which causes the movement of the push pin; b) the drive assembly includes an electric motor, a gear assembly configured to be driven by the electric motor, an inclined plate rotatably connected to the gear assembly and can be driven by the gear assembly, and a wobble plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; c) the wobble plate is configured to not rotate with the inclined plate; d) an anti-rotational assembly is connected between the wobble plate and a housing containing the bore; e) the anti-rotational assembly can limit rotation of the wobble plate and permit axial movement of the wobble plate; f) the anti-rotational assembly includes at least one spring that extends axially between the wobble plate and the housing; g) the anti-rotational assembly includes a pin that extends radially between the wobble plate and the housing; h) the

outer perimeter of the wobble plate includes a series of teeth and grooves; and/or i) the pin is configured to fit in a groove between adjacent teeth.

(19) In some forms: a) an arc angle of the first end is less than 180 degrees; and/or b) the bore, the piston, and the push pin can be such that a movement of the push pin relative to the bore causes translation of the piston within the bore to the extended position along the translation axis.

(20) A reciprocating assembly for driving a working portion of a tool, the reciprocating assembly including: a bore configured to contain hydraulic fluid; a piston received within the bore and configured to move within the bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the piston within the bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the bore; and a push pin contacting the piston; a drive assembly configured to supply a force which causes the movement of the push pin, the drive assembly including a non-rotational plate configured to supply the force by rocking between a first plate position and a second plate position, the plate being closer to the piston in the first plate position than in the second position; wherein the bore, the piston, and the push pin are configured such that a movement of the push pin relative to the bore causes translation of the piston within the bore to the extended position along a translation axis; and wherein the push pin is configured to pivot during the movement of the push pin, between an in-line position and an angled position, the in-line position substantially parallel to the translation axis, the push pin being in the in-lined position when the plate is in the first plate position and the push pin being in the angled position in the second plate position.

(21) In some forms: a) the piston includes an internal cavity extending at least partially a length of the piston; b) the push pin is at least partially received within the internal cavity; c) the internal cavity includes a terminal end and an open end; and/or d) the open end being wider than the terminal end.

(22) In some forms: a) the push pin includes a first concave region; b) the plate includes a second concave region; c) a ball bearing at least partially received within the first concave region and in the second concave region; d) the ball bearing configured to permit relative movement between the push pin and the plate; and/or e) the second concave region has a larger radius of curvature than the first concave region.

(23) In some forms: a) the push pin includes a hyperboloid shape; b) the push pin further includes a first end having a first concave region and a second end having a second concave region; c) a first ball bearing is received within the first concave region and contacts the piston; d) a second ball bearing is received within the second concave region and contacts the plate; e) the push pin further includes a first end having a first hemispherically shaped region and a second end having a second hemispherically shaped region; f) the first hemispherically shaped region contacts the piston; g) the second hemispherically shaped region contacts the plate; h) an O-ring connected to an outer surface of the push pin; i) the O-ring is compressible.

(24) In some forms: a) the bore is a first bore, the push pin is a first push pin and a piston is a first piston; b) a second bore is spaced apart from the first bore and can contain hydraulic fluid; c) a second piston received within the second bore and can move within the second bore in a reciprocating motion between an extended position and a retracted position; d) displacement of the second piston within the second bore can cause displacement of hydraulic fluid when hydraulic fluid is in the second bore; e) the drive assembly can supply a force which causes the movement of the second push pin; f) the plate is closer to the first piston in the first plate position and closer to the second piston in the second position; g) the first bore is parallel to the second bore; h) the translational axis is a first translational axis; i) the second piston moves along a second translational axis that is parallel to the first translational axis; j) the second push pin can pivot during the movement of the second push pin, between a second in-line position and a second angled position; k) the second in-line position substantially parallel to the second translation axis; l) the second push pin being in the second in-lined position when the plate is in the second plate position and the

second push pin being in the angled position in the first plate position; m) a second push pin contacting the second piston.

(25) In some forms: a) the drive assembly includes an electric motor, a gear assembly configured to be driven by the electric motor, an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly, and the non-rotational plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; and/or b) the non-rotational plate is configured to not rotate with the inclined plate.

(26) A push pin for use in a reciprocating assembly for driving a working portion of a tool, the push pin including: a body having a first end and a second end opposite to the first end, wherein, a first cutout formed at the first end, and a second cutout formed at the second end; wherein the first end of the body is configured to be received within a bore of a piston such that movement of the push pin relative to the bore causes translation of the piston within the bore to the extended position along a translation axis.

(27) In some forms: a) the body has a substantially cylindrical shape; b) the push pin includes a hyperboloid shape; c) an arc angle of the first cutout is less than 180 degrees; d) the push pin is constructed from a rigid material and can be incompressible when contacting the piston; and/or e) the first cutout is configured to receive a first ball bearing and the second cutout can receive a second ball bearing.

(28) A tool including: a working portion; a piston configured to move in a reciprocating motion to drive a hydraulic fluid and drive the working portion, the piston including an internal cavity extending at least partially along a length of the piston; and a push pin received within the internal cavity of the piston; wherein the piston and the push pin are configured such that a movement of the push pin drives movement of the piston along a translation axis; and wherein the push pin is configured to pivot during the movement of the push pin, between an in-line position and an angled position, the in-line position substantially parallel to the translation axis.

(29) In some forms: a) the cavity extends along a majority of the length of the piston; and/or b) the working portion includes a first jaw and a second jaw movable relative to the first jaw to perform a crimping function.

(30) In some forms: a) a drive assembly for driving movement of the push pin; b) a drive assembly includes an electric motor, a gear assembly configured to be driven by the electric motor, an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly, and a wobble plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; c) the wobble plate is configured to not rotate with the inclined plate; and/or d) movement of the wobble plate drives the movement of the push pin.

(31) In some forms: a) the piston is a first piston, the push pin is a first push pin, and the translational axis is a first translational axis; b) a second piston can move in a reciprocating motion to drive hydraulic fluid and drive the working portion; c) the second piston including a second internal cavity extending at least partially along a length of the second piston; d) a second push pin received within the second internal cavity of the second piston; e) the second piston and the second push pin allow a movement of the second push pin drives movement of the second piston along a second translation axis that is parallel to the first translational axis; f) the second push pin can pivot during the movement of the second push pin, between a second in-line position and a second angled position, the second in-line position substantially parallel to the second translation axis; and/or g) an axis along the first push pin in the angled position is obliquely oriented to an axis along the second push pin in the second angled position.

(32) In some forms: a) the piston includes a first end with a first cutout having the first radius of curvature; b) the first cutout at least partially receives a first ball bearing; c) the first end and the first ball bearing being received within the internal cavity; and/or d) the push pin is configured to rotate about the first ball bearing.

(33) In some forms: a) the push pin includes two opposite hemispherically shaped ends; b) one of

the hemispherically shaped ends is received within the internal cavity and the other of the hemispherically shaped ends is coupled to a drive assembly that can supply a force which causes the movement of the push pin; b) the push pin includes a hyperboloid shape; c) an O-ring connected to an outer surface of the push pin; and/or d) the O-ring is compressible.

(34) A tool including: an electric motor; a gear assembly configured to be driven by the electric motor; an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly; and a wobble plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; an anti-rotational assembly configured to prevent the wobble plate from rotating with the inclined plate; and a push pin driven by movement of the wobble plate between a first position and a second position, the push pin configured to move a piston along a piston axis to drive a hydraulic fluid in the first position; wherein the push pin is parallel to the axis in the first position and is oblique to the piston axis in the second position.

(35) In some forms: a) the anti-rotational assembly includes at least one spring that extends axially between the wobble plate and the housing; b) the spring is compressible along a spring axis parallel to the piston axis; c) the anti-rotational assembly includes a pin that extends radially between the wobble plate and the housing; d) the outer perimeter of the wobble plate includes a series of teeth and grooves; e) the pin can fit in a groove between adjacent teeth; and/or f) each tooth of the series of teeth are frustoconically shaped.

(36) According to another aspect of various examples of the present disclosure, there is provided a reciprocating assembly of any of the previous aspects; and a working portion configured to be driven by the reciprocating assembly.

(37) The disclosure herein should become evident to a person of ordinary skill in the art given the following enabling description and drawings. The drawings are for illustration purposes only and are not drawn to scale unless otherwise indicated. The drawings are not intended to limit the scope of the invention. The following enabling disclosure is directed to one of ordinary skill in the art and presupposes that those aspects within the ability of the ordinarily skilled artisan are understood and appreciated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects and advantageous features of the present disclosure will become more apparent to those of ordinary skill when described in the detailed description of preferred examples and reference to the accompany drawings.

(2) FIG. 1 shows a tool according to an exemplary example of the disclosure.

(3) FIG. 2 shows a perspective view of a portion of the tool of FIG. 1.

(4) FIG. 3 shows an exploded view of the tool of FIG. 2.

(5) FIG. 3A shows a detailed exploded view of a reciprocating assembly of the tool of FIG. 1.

(6) FIG. 4 shows a cross-sectional view of the tool of FIG. 1, illustrating the tool in a first position.

(7) FIG. 5 shows a detailed view of the cross-sectional view of FIG. 4.

(8) FIG. 6 shows a detailed view of an alternative version of the cross-sectional view of FIG. 4.

(9) FIG. 7 shows a cross-sectional view of the tool of FIG. 1, illustrating the tool in a second position.

(10) FIG. 8 shows a detailed view of the cross-sectional view of FIG. 7.

(11) FIG. 9 shows a cross-sectional view of a piston used with the tool of FIG. 1.

(12) FIG. 10 shows a cross-sectional view of a push pin used with the tool of FIG. 1.

(13) FIG. 11 shows a cross-sectional view of an alternate example of a push pin.

(14) FIG. 12 shows a cross-sectional view of a further alternate example of a push pin.

(15) FIG. 13 shows a detailed view of a cross-sectional view of the tool illustrating springs of an

anti-rotational assembly.

(16) FIG. **14** shows a front view of an alternate wobble plate.

(17) FIG. **15** shows a side view of the alternate view of the wobble plate of FIG. **14**.

(18) FIG. **16** shows a partially exploded view of an alternate anti-rotational assembly including the wobble plate of FIG. **14**.

(19) FIG. **17** shows a perspective view of the alternate anti-rotational assembly of FIG. **16**.

DETAILED DESCRIPTION

(20) FIG. **1** illustrates a tool **100**. The illustrated tool **100** is a handheld power tool, which allows the tool **100** to be portably used at different work sites. The tool **100** may be powered by a rechargeable battery **110** (e.g., an 18V battery pack). This type of battery **110** may be interchanged with other types of power tools.

(21) The tool **100** includes an elongated body **120** that a user grips while using the tool **100**. The elongated body **120** includes one or more controls **130** (e.g., two shown) for operating the tool **100**. The battery **110** is removably connected to the one end of the body **120**. The one or more controls **130** may be disposed between the ends of the elongated body **120**.

(22) The illustrated tool **100** includes a working portion **140** connecting to an opposite end of the elongated body **120** from the battery **110**. The illustrated working portion **140** is configured for performing a crimping or cutting function. The working portion **140** includes a pair of jaws **145** that move relative to one another to cut or crimp a piece of material. Other examples of tools (not shown) may include other working portions **140**.

(23) The illustrated tool **100** may be a hydraulic tool (e.g., an in-line hydraulic tool). As will be described in more detail below, the electrical energy from the battery **110** may be used to drive a hydraulic fluid, that in turn drives the working portion **140**.

(24) As shown in FIGS. **2** to **8**, the tool **100** may include a motor **200** (e.g., a brushless DC motor). The motor **200** may be electrically connected to the battery **110** when the battery **110** is connected to the body **120**. When the tool **100** is powered on, current from the battery **110** may drive operation of the motor **200**.

(25) With continued reference to FIGS. **3** and **4**, the tool **100** may include a gear assembly **210** that is received within a transfer casing or housing **205** and that is connected to the motor **200**. Rotation of the motor **200** may drive the rotation of the gear assembly **210**.

(26) As shown in FIGS. **4** to **8**, the tool **100** may also include an inclined plate **220** within the housing **205** that is connected to the gear assembly **210**. For example, the gear assembly **210** may be disposed between the motor **200** and the inclined plate **220**. The gear assembly **210** may receive energy from the motor **200** and may output energy to the inclined plate **220**. The illustrated example of the tool **100** shows the motor **200**, the gear assembly **210**, and the inclined plate **220** in-line (e.g., along a common axis), although other orientations may be used. When the motor **200**, the gear assembly **210**, and the inclined plate **220** are oriented in-line, the common axis may extend along the elongated body **120** (e.g., between the ends of the elongated body **120**).

(27) In the illustrated example, the inclined plate **220** may include a first planar surface **222** and a second planar surface **224**. The first planar surface **222** may be disposed proximate to the gear assembly **210** and the second planar surface **224** may be disposed opposite to the first planar surface **222**. Each of the planar surfaces **222**, **224** may be relatively flat (e.g., not curved), although other structures may be possible.

(28) The second planar surface **224** may be oblique with respect to the first planar surface **222**. For example, the second planar surface **224** may be oriented so that the distance between the first and second surfaces **222**, **224** is greater at one side than at another side. The second planar surface **224** may also be oriented obliquely with respect to an axis about which the motor **200** rotates.

(29) A wobble plate **230** is illustrated as being in contact with the second planar surface **224** and disposed within the housing **205**. The wobble plate **230** may include a third planar surface **232** that rests against the second planar surface **224**. For example, the third planar surface **232** may be

oriented so that it is parallel to the second planar surface **224**.

(30) An opposite end of the wobble plate **230** from the third planar surface **232** includes a surface with a series of grooves. For example, the wobble plate **230** may include a central groove **234** and side grooves. For example, the wobble plate **230** may include four side grooves, although any number (e.g., three, five, six, etc.) of grooves may be included.

(31) In some examples, the grooves may include a pair of first grooves **236** and a pair of second grooves **237**. In the illustrated example, the first and second grooves **236**, **237** may be substantially the same size, although in other examples the one of the pairs of grooves may be larger than the other. All the grooves **234**, **236**, **237** may be spaced apart from one another. For example, the first and second grooves **236**, **237** may be spaced (e.g., equally spaced) around the perimeter of the wobble plate **230** and the central groove **234** may be spaced in the center of the first and second grooves **236**, **237**. Each first groove **236** may be disposed approximately 180° apart from one another, and each second groove **237** may be disposed approximately 180° apart from one another.

(32) In the illustrated example, each of the grooves **234**, **236**, **237** may have a rounded shape. In some examples, each groove **234**, **236**, **237** may have a constant curvature, although in other examples at least one of the grooves **234**, **236** may include a non-uniform curvature. Additionally, the angle of curvature in each of the grooves **234**, **236**, **237** may be less than about 180°, although some examples may include a greater angle of curvature. For example, the illustrated grooves **234**, **236**, **237** may each have an angle of curvature that is less than about 90°.

(33) A hydraulic assembly **300** is connected to the housing **205**. For example, the hydraulic assembly **300** may generally be positioned outside of the housing **205** although at least some of the hydraulic assembly **300** may be at least partially received within the housing **205**. As will be described in more detail below, the hydraulic assembly **300** includes pistons for driving a working fluid that ultimately drives operation of the working portion **140**.

(34) As shown in FIGS. 2 and 3, the hydraulic assembly **300** includes a housing **305**. The illustrated housing **305** is partially received within the housing **205**. The housing **305** of the hydraulic assembly **300** includes a pair of inlet ports **310** (see e.g., FIG. 4) that are received within a reservoir **307** that can store hydraulic fluid (e.g., oil). The reservoir **307** may be permanently or removably connected to the housing **305**.

(35) As shown in FIG. 4, each inlet port **310** extends into a bore **315** within the housing **305**. The bore **315** may be open at an opposite end from the respective inlet port **310**, with the opening of the bore **315** being disposed within the housing **205**. A wall **320** may separate the bores **315** from one another. Each of the bores **315** may contain hydraulic fluid during operation of the tool **100**.

(36) As shown in FIGS. 4, 6, and 8, piston **325**, a spring **330**, and a valve **335** are located within each bore **315**. A valve **335** is positioned within each bore **315** proximate to the respective inlet port **310**. The valve **335** may include a passage **337** that can permit fluid to travel from the respective inlet **310** to the respective bore **315**. The valve **335** may be a one-way valve so that fluid from the reservoir **307** can enter the passageway **337** to reach one of the bores **315**, while fluid within the bores **315** cannot return to the reservoir **307** through the passageway **337**. The spring **330** is connected to the bore proximate to the respective inlet **310** and the piston **325** is connected to the other end of the spring **330** and extends toward the other end of the bore **315** proximate to the housing **205**.

(37) The spring **330** may be biased toward the wobble plate **230** and may be compressible as the respective piston **325** moves toward the inlet **310**. Each spring **330** therefore may oppose the translational motion of the respective piston **325** in one direction.

(38) As shown in FIG. 9, each piston **325** includes a first end **340** and a second end **345** opposite to the first end **340**. In the illustrated example, the first end **340** has a solid surface. For example, the first end **340** may include a planar surface, which may be substantially perpendicular to the direction of travel of the piston **325**.

(39) In some forms, the first end **340** may extend away from the body **350** of the piston **325**. For

example, the first end **340** may project beyond the body **350** and may have a different width than the remainder of the body **350**. In the illustrated example, the first end **340** may be at least partially cylindrical with a smaller diameter than the diameter of the body **350**.

(40) In certain forms, an undercut **351** may be positioned between the first end **340** and the remainder of the body **350**. The undercut **351** may be formed because of the smaller diameter of the first end **351**. An end of the spring **330** may be coupled to the body **350** via the undercut **351**. In other examples, a seal may be disposed at least partially within the undercut **351**.

(41) The second end **345** may have an opening that extends into a cavity **355** of the body **350**. The cavity **355** may extend at least partially along the length of the body **350**. For example, the illustrated example in FIG. **9** shows the cavity **355** extending along a majority of the length of the body **350** but does not reach the first end **340**. Although other lengths are possible in other examples (e.g., extending to the first end **340**, extending less than a majority of the length of the body **350**, etc.).

(42) In the illustrated example, the cavity **355** may be substantially cylindrical and may include a terminal end **360** that is at least partially spherical in shape. For example, the terminal end **360** may have a hemispherical shape, although other shapes may also be used.

(43) The open end **365** of the cavity **355** (e.g., proximate to the second end **345** and opposite from the terminal end **360**) forms the opening of the second end **345** and may have a different width than the remainder of the cavity **355**. For example, the open end **365** of the cavity **355** may have a frustoconical shape and may taper outwardly so that the entrance to the cavity **355** at the second end **345** is wider than the width of the cavity **355** along the remainder of the cavity **355**. The outer dimension of the piston **325** may remain substantially uniform along the length of the body **350**. The thickness of a wall of the body **350** (e.g., distance between the wall of the cavity **355** and the outer wall of the piston **325**) may therefore be less proximate to the second end **345** than to the first end **340**. In other examples, the cavity **355** may have a substantially uniform width along its entire length.

(44) Returning to FIGS. **4** to **8**, a push pin **370** may be positioned within each cavity **355**. The push pin **370** may be a substantially cylindrical member. Each end of the push pin **370** may include a cutout **375** with a rounded or partially spherical shape. As better shown in FIG. **10**, each cutout **375** may be less than hemispherical, although any shaped cutout **375** may be used.

(45) Returning to FIGS. **4** to **8**, a spherical member (e.g., a ball bearing) **380** may be at least partially received with each of the cutouts **375**. In other words, one ball bearing **380** may be partially received within each of the cutouts **375** on each of the push pins **370**. Because each cutout **375** has an arc angle of less than 180° (i.e., is not fully hemispheric), each cutout **375** may cradle the respective ball bearing **380** without fully surrounding it.

(46) The ball bearings **380** may be a substantially rigid member that may not be substantially compressed while in use. The ball bearing **380** may be constructed from a metal (e.g., steel), a ceramic, a plastic, or a similar material.

(47) The ball bearings **380** additionally rest on another surface. For example, one ball bearing **380** is received entirely within the cavity **355** and is positioned at the terminal end **360**. The hemispherical shape of the terminal end **360** may substantially match the shape of the ball bearing **380** (e.g., it may have the substantially the same radius of curvature). The ball bearing **380** therefore may be substantially flush against the surface of the terminal end **360**.

(48) The push pin **370** extends away from the terminal end **360** toward the second end **345** of the body **350**. In the illustrated example, the push pin **370** may extend into the open end **365** (e.g., the frustoconical portion) but may not extend beyond the second end **345**. However, other examples may include a longer push pin **370** that extends beyond the second end **345** and into the housing **205**.

(49) The ball bearing **380** proximate to the open end **365** may be received between a cutout **375** of the piston **325** and a respective one of the side grooves **236**. The ball bearing **380** therefore may be

positioned at least partially outside of the cavity **355**. The cutout **375** of the push pin **370** may have a substantially similar shape as the ball bearing **380** (e.g., substantially the same radius of curvature). Each first side groove **236** may have a larger radius of curvature as compared to the ball bearing **380** so that the ball bearing **380** is loosely received within the respective first side groove **236**.

(50) In some forms, the ball bearings **380** may be substantially the same size and the cutouts **375** in the push pins **370** may also be substantially the same size as one another. The push pins **370** may be used with either end inserted into the cavity **355**.

(51) The wall **320** may also include a cutout **385**. A central ball bearing **390** may be received at least partially within the cutout **385** and the central groove **234** of the wobble plate **230**. In the illustrated example, the central ball bearing **390** may be larger than the ball bearings **380**. Accordingly, the cutout **385** of the wall **320** may be larger than the cutouts **375** of the push pin **370**. The central ball bearing **390** and the central groove **234** may have substantially the same radius of curvature so that the central ball bearing **390** may be substantially flush with the surface of the central groove **234**.

(52) As shown in FIG. **11**, an alternate form of the push pin **400** may have a different outer shape than the push pin **370**. For example, instead of a rectangular cross section (e.g., forming a cylindrical body) like in the push pin **370** in FIG. **10**, the push pin **400** may have an hourglass shaped body as viewed in cross section. In other words, a center of the push pin **400** may be narrower than the ends of the push pin **400**. Additionally, the outer surface of the push pin **400** may curve between the ends. When viewed in three dimensions, this shape may form a hyperboloid. In the illustrated example, the widest portion of the push pin **400** may be approximately the same width as the widest portion of the push pin **370**.

(53) In other examples, the outer surface of the push pin **400** may not form a smooth curve, and may instead include an angled portion (e.g., where the narrowest point of the push pin **400** may be formed at a point). When viewed in three dimensions, the push pin **400** may not be rounded and may include a number of sides to approximate a curved surface.

(54) The push pin **400** may include cutouts **405** at either end. The cutouts **405** may be substantially the same size as the cutouts **375** and may receive the same sized ball bearings **380**. The push pin **400** therefore may be interchanged with the push pin **370**.

(55) As shown in FIG. **12**, a further alternate form of the push pin **450** may have an outer shape similar to the push pin **370**. For example, the push pin **450** may have a substantially cylindrical body and appear rectangular in cross section. Instead of cutouts at either end, the push pin **450** may include rounded (e.g., semi-spherical) ends **455**. A tool **100** that uses the push pin **450** may not use the ball bearings **380** (e.g., the ends **455** may be similarly shaped to the ball bearings **380**). This may simplify manufacturing and/or assembly as fewer parts are needed. In other examples (not shown), the push pin **450** may include the hourglass shape like the push pin **400**, while still including the rounded ends **455**.

(56) In still other examples, the body of the push pin may not be symmetrical along its elongated axis. For example, the curved portion between either end of the push pin **400** may be included on a single side while the other may be substantially straight (in cross section).

(57) Returning to FIGS. **4** and **7**, an outlet **410** may be disposed within the wall **320**. The outlet **410** may therefore be disposed between each of the bores **315**. Although not shown, the outlet **410** may be fluidly connected to each of the bores **315**.

(58) The outlet **410** may terminate at a chamber **415**, thus providing fluid communication between the bores **315** and the chamber **415**. A valve **420** may be disposed within the outlet **410** to control fluid flow between the bores **315** and the chamber **415**.

(59) Although not shown, the chamber **415** may include a piston, which may be driven by fluid flowing from the bores **315**. The piston in the chamber **415** may be capable of extending and retracting to move the working portion **140** of the tool **100**.

(60) In use, the user may connect the battery **110** to the body **120** and actuate the tool **100** by engaging at least one of the controls **130**. This causes electrical energy to power the motor **200**, which in turn rotates the gear assembly **210**. The inclined plate **220** is connected to the gear assembly **210** and is driven by the rotation of the gear assembly **210**.

(61) The rotation of the inclined plate **220** causes a change in orientation of the inclined plate **220**. In other words, the rotational position of the second planar surface **224** changes as the inclined plate rotates **220**. For example, FIGS. **4** to **6** illustrates the inclined plate **220** with the second planar surface **224** oriented so that the greater width relative to the first planar surface **222** is at the top of the figure. In FIGS. **7** and **8**, the inclined plate **220** has rotated approximately 180° so that the second planar surface **224** is oriented so that the greater width relative to the first planar surface **222** is at the bottom of the figure. Rotation between these locations may continue as the inclined plate **220** continues to be driven by the motor **200** via the gear assembly **210**. For example, the orientation of the second planar surface **224** may oscillate between these positions as the motor **200** drives the gear assembly **210**.

(62) The wobble plate **230** contacts the second planar surface **224** as the inclined plate **220** rotates. However, unlike the inclined plate **220**, the wobble plate **230** does not rotate with the inclined plate **220**. Instead, the rotation of the inclined plate **220** causes the wobble plate **230** to rock or wobble, although some examples may include a wobble plate **230** that moves with some rotation.

(63) As shown in FIGS. **3A** and **13**, one or more springs **238** (e.g., two shown) are used to limit rotation of the wobble plate **230**, e.g., form an anti-rotational assembly. The springs **238** may be positioned within the second grooves **237** of the wobble plate **230** and may each extend into a pocket in the housing **305**. In use, the axial load from the springs **238** may be greater than the bearing friction from any or all the bearings **380**, **390**. This may sufficiently restrain rotational movement while permitting axial movement in the form of the reciprocating wobble. As this movement occurs, the compression in the springs **238** may change. In other words, the top spring **238** and the bottom spring **238** (as viewed in FIG. **13**) may alternatively have a different compressive load.

(64) As shown in FIGS. **14** to **17**, an alternate example may use a pin arrangement in an anti-rotational assembly to limit rotational movement of the wobble plate **230**. In the illustrated example, the outer perimeter of the wobble plate **230** may include a series of teeth **248** and grooves **250**. The teeth **248** and grooves **250** may have a partially stepped pattern, although other patterns may be used (e.g., inclined, curved, etc.). For example, the illustrated teeth **248** may slightly narrow in a direction distal to the surface of the wobble plate **230**. The housing **205** may include one or more holes **255**, which may align with one of the grooves **250** when the tool **100** is assembled. A pin **260** may be inserted through a hole **255** and aligned with one of the grooves **250**. The pin **260** may have approximately the same width as the width of the groove **250**. For example, the pins **260** may have a frustoconical shape that corresponds to the shape of the teeth **248** and may contact the surface of the teeth **248** when fully inserted in the respective hole **255**. Thus, the pin **260** may fit between adjacent teeth **248** but there may not be substantially room for the pin **260** to shift between the adjacent teeth **248**. In use, the size of the pin **260** between the teeth **248** may limit rotational movement of the wobble plate **230**. However, because the teeth **248** are disposed only along the radial direction, the wobble plate **230** may still be movable in the axial direction.

(65) In some forms, the pin **260** may be a separate element and may be inserted through a hole **255** (see e.g., FIG. **2**) after the wobble plate **230** is assembled. Any number of pins **260** may be used (e.g., one, two, three, four, etc.). In other forms, the pins **260** may be permanently connected to the housing **205** and may assist in guiding the wobble plate **230** into position during assembly. In still other example, the pin **260** may be connected to the wobble plate **230**. The pin **260** may retract (e.g., against the bias of a spring) as the wobble plate **230** is inserted into the housing **205** during assembly. Once aligned with the hole **255**, the pin **260** may extend through the hole **255**. In this example, the wobble plate **230** may not include the teeth **248** and grooves **250** and may instead

include an elongated hole **255** in the axial direction.

(66) In some forms, the pin **260** and the springs **238** may operate with the same wobble plate **230**.

(67) As described above, the rotation of the inclined plate **220** changes the position of the second planar surface **224**. Because the third planar surface **232** of the wobble plate **230** is flush or substantially flush with the second planar surface **224**, the inclination of the wobble plate **230** changes because of the rotation of the inclined plate **220**.

(68) For example, the wobble plate **230** is closer to the upper bore **315** as shown in FIGS. **4** to **6** and is closer to the lower bore **315** as shown in FIGS. **7** and **8**. The wobble plate **230** may rock between these two positions as the inclined plate **220** continues to rotate.

(69) As the wobble plate **230** rocks between the positions illustrated in FIGS. **4** and **7**, the wobble plate **230** pivots the push pins **370** within the respective cavity **355**. For example, a ball bearing **380** is sandwiched between the terminal end **360** of each cavity **355** and the cutout **375** of each push pin **370**. The spherical ball bearing **380** allows for pivoting movement of the respective push pin **370** because the rounded cutout **375** can move along the surface of the respective ball bearing **380**. Similarly, on the other end, the additional ball bearing **380** is received between the opposite cutout **375** and the respective first side groove **236**. The larger radius of curvature of the first side groove **236** as compared to the ball bearing **380** allows the wobble plate **230** to move between the positions of FIGS. **4** and **7**.

(70) Additionally, the central ball bearing **390** is received between the wall **320** and the central groove **234**. The central ball bearing **390** creates contact between the wall **320** and the wobble plate **230**. The central ball bearing **390** may also provide a pivot point for the wobble plate **230** to rock against as the inclined plate **220** rotates.

(71) The pivoted position of each push pin **370** may correspond with a position of the piston **325**. Starting with FIGS. **4** to **6**, the tool **100** is illustrated so that the inclined plate **220** is oriented with the wider portion (distance between the first and second planar surfaces **222**, **224**) on top. The wobble plate **230** is similarly oriented so that the top first side groove **236** is closer to the respective piston **325** than the bottom first side groove **236** is to the respective piston **325**. In this position, the top push pin **370** is oriented substantially along the axis **367** of the respective piston **325** (e.g., between the terminal end **360** and the open end **365**). The bottom push pin **370** may be angled relative to the axis **367** of the bottom piston **325**. For example, the bottom push pin **370** may pivot toward the top push pin **370** in this position.

(72) In FIGS. **7** and **8**, the tool **100** is illustrated so that the inclined plate **220** is oriented with the wider portion on the bottom. The wobble plate **230** is similarly oriented so that the bottom first side groove **236** is closer to the respective piston **325** than the top first side groove **236** is to the respective piston **325**. In this position, the bottom push pin **370** is oriented substantially along the axis **367** of the respective piston **325** (e.g., between the terminal end **360** and the open end **365**). The top push pin **370** may be angled relative to the axis **367** of the top piston **325**. For example, the top push pin **370** may pivot toward the bottom push pin **370** in this position.

(73) In the illustrated examples, the width of each push pin **370** may be less than the width of the respective cavity **355**. More particularly, the width of each cavity **355** may be wide enough to provide clearance for the above-described pivoting movement of a push pin **370**. For example, the bottom push pin **370** in FIGS. **4** to **6** and the top push pin **370** in FIGS. **7** and **8** may avoid contact with the wall of the cavity **355** in the pivoted position. This may assist in limiting frictional wear between the push pins **370** and the cavity **355** to better maintain the structural integrity of the push pins **370**.

(74) As shown in FIGS. **5** and **6**, some forms of the tool **100** may include one or more O-rings. In one form as shown in FIG. **5**, each piston **325** includes a first O-ring **372** proximate to the respective open end **365**. Each push pin **370** may move relative to the respective first O-ring **372** and may contact the respective first O-ring **372** proximate to its end of travel. In another form as shown in FIG. **6**, a second O-ring **374** may be coupled to an outer surface of one or more of the

push pins **370**. The second O-ring **374** moves with the respective push pin **370** so that there is no relative movement. This second O-ring **374** may be positioned proximate to a middle of the respective push pin **370** and may contact the wall of the cavity **355** between the terminal end **360** and the open end **365**. As shown in FIG. **6**, both the first and second O-rings **372**, **374** may be included in the same tool **100**, although other examples may include only the second O-ring **374** connected to the push pin **370**.

(75) The O-ring **374** may assist with aligning the push pin **370** within the respective cavity **355** along the axis **367**. For example, the outer width of the O-ring may be substantially equal to the width of the cavity **355**. This may minimize or eliminate any clearance and help to ensure that the push pin **370** is oriented along the axis **367**.

(76) In certain forms, the O-rings **372**, **374** may be compressible. For example, the push pin **370** may compress the O-rings **372**, **374** against the wall of the cavity **355** as the push pin **370** pivots. The O-rings **372**, **374** therefore may not limit motion of the push pin **370**.

(77) The frustoconical shape of the open end **365** of each cavity **355** may provide additional clearance for the respective push pin **370** to assist in better avoiding contact. The frustoconical shape with the widest portion proximate to the second end **345** provides additional space for the push pin **370** to move and increases its range of travel.

(78) In forms that include the O-ring **372** proximate to the open end **365**, the O-ring **372** may provide additional assistance in minimizing contact between the push pin **370** and the wall of the cavity **355**. For example, the push pins **370** may contact the compressible O-ring **372** and limit the increased frictional engagement with the wall of the cavity **355**.

(79) The hourglass or hyperboloid shape of the push pin **400** in FIG. **11** may provide additional clearance between the push pin **400** and the wall of the cavity **355**. For example, the hourglass shaped push pin **400** includes portions with a reduced width (as compared to the push pin **370**). This reduced width may provide additional spacing between the cavity **355** to account for tolerances in the width of the cavity **355**. For example, the push pin **400** may not contact the O-rings **372** in FIG. **5** at the end of travel.

(80) Returning to FIGS. **4** to **8**, the pivoting movement of the push pins **370** corresponds to movement of the pistons **325**. The pistons **325** are designed to move linearly within the respective bore **315**. Movement of the push pins **370** because of the changing position of the wobble plate **230** affects the translational position of the pistons **325**.

(81) As illustrated in FIGS. **4** to **6**, the top piston **325** is closer to the respective valve **335** than the bottom piston **325** is to the respective valve **335**. In other words, the top spring **330** is more compressed than the bottom spring **330**, and there is less space between the first end **340** and the valve **335** in the top bore **315** than in the bottom bore **315**.

(82) As described above, the top first side groove **236** is proximate to the top bore **315** and the top push pin **370** is substantially straight. In this position, the push pin **370** may be providing a pushing force to the piston **325** that overcomes the bias of the spring **330**. In other words, as the inclined plate **220** turns and the wobble plate **230** rocks to its position in FIGS. **4** to **6**, the rigid body of the top push pin **370** may be pushed toward the terminal end **360** of the piston **325**, thus driving the piston **325** toward the valve **335**.

(83) Simultaneously, the lower first side groove **236** is distal to the bottom bore **315** and the bottom push pin **370** is inclined relative to the axis **367** along the cavity **355** of the bottom piston **325**. The movement of the lower first side groove **236** away from the bore **315** provides a limited pushing force via the push pin **370** that is not sufficient to overcome the bias of the spring **330**.

(84) As the inclined plate **220** rotates to the position in FIGS. **7** and **8**, the lower first side groove **236** moves toward the bottom bore **315** and the bottom push pin **370** is substantially straight. As the wobble plate **230** rocks toward this position, the push pin **370** is pushed toward the terminal end **360** of the piston **325**. This pushing force overcomes the bias of the spring **330** and drives the piston **325** toward the valve **335**.

(85) Simultaneously, the upper first side groove **236** moves away from the top bore **315** and the top push pin **370** becomes inclined relative to the axis **367** along the cavity **355** of the top piston **325**. The movement of the top first side groove **236** away from the bore **315** reduces the pushing force applied toward the terminal end **360** in FIGS. **4** to **6** and allows the bias of the spring **330** to overcome any force from the top push pin **370**. This allows the top piston **325** to move toward a retracted position in FIGS. **7** and **8**.

(86) This movement of the pistons **325** drives the movement of the hydraulic fluid during operation of the tool **100**. More specifically, displacement of each piston **325** causes displacement of the hydraulic fluid either into or out of the bore **315**. For example, in the retracted position (e.g., the bottom piston **325** in FIGS. **4** to **6** and the top piston **325** in FIGS. **7** and **8**), space within the bore **315** increases. The respective valve **335** may allow hydraulic fluid to flow into the bore **315** from the reservoir via the inlet **310** (e.g., so that the bore **315** contains the hydraulic fluid). As the piston **325** moves toward the extended position (e.g., the top piston **325** in FIGS. **4** to **6** and the bottom piston **325** in FIGS. **7** and **8**), the fluid within the bore **315** is driven out. In the extended position, the valve **335** is closed to limit backflow into the reservoir. Instead, a valve (e.g., valve **420**) may be opened to allow the fluid to leave the bore **315** and travel toward the chamber **415**. An O-ring **417** may be disposed around an outer perimeter of each piston **325** in order to limit flow out of the bore **315** and toward the motor **200**.

(87) The continuing rotation of the inclined plate **220** (e.g., if the user continues to actuate the control **130**) continues to drive the movement of the pistons **325** in a reciprocating motion so that the hydraulic fluid continues to be driven from the reservoir and into the chamber **415**. Once in the chamber **415**, another piston (not show) may continue to drive the fluid toward the working portion **140** to actuate the jaws **145**.

(88) The efficiency on the fluid moving process (e.g., drawing fluid into the bore **315** and pushing the fluid toward the chamber **415**) may be a product of the linear force provided by the piston **325**. More specifically, the pistons **325** are intended to move in the left-right direction as shown in FIGS. **4** to **8**. Particularly in the extended position of the respective piston **325**, movement directly along this axis **367** will provide the most force for pushing against the spring **330**.

(89) To limit losses (e.g., via friction) because of the inclination of the wobble plate **230**, the pivoting movement of the push pins **370** may help to direct more of the force along the axis **367** of the respective piston **325**.

(90) As described above, each push pin **370** can pivot because of contact with the ball bearings **380** and the clearance within the cavity **355**. This movement is specifically driven by the movement of the wobble plate **230**. Because of the inclination of the wobble plate **230** relative to the bore **315**, the wobble plate **230** may provide a force that is oriented in two directions (e.g., a component may be oriented in the vertical and the horizontal direction as viewed in FIGS. **4** to **8**). While the horizontal component is desired to move the piston **325** against the spring, the vertical component may move the piston **325** into the wall of the bore **315**. This may cause frictional engagement while moving the piston **325**, so that a greater resultant force is required to overcome the spring bias and the outer surface of the piston **325** may wear.

(91) The push pins **370** can help address this problem by tending to reduce the vertical component of the force and increase the horizontal component so that a greater percentage of the resultant force is directed along the axis **367** of the cavity **355**. The force is transferred along the substantially rigid push pin **370** and is applied to the respective piston **325** proximate to the first end **340** (e.g., proximate to the terminal end **360**) as opposite to proximate to the second end **345**. Moving the force application point toward the first end **340** may enable the vertical force component to be lessened.

(92) The ball bearing **380** between the first side groove **236** and the push pin **370** may be maintained substantially in the center of the respective first side groove **236**. In the extended position, the center of the first side groove **236** is closest to the second end **345**, and more

particularly is closest to the axis 367 of the cavity 355. Because the push pin 370 is substantially rigid (e.g., does not substantially stretch or compress) the shortest distance between the terminal end 360 and the center of the first side groove 236 is a substantially straight line along the axis 367. The push pin 370 therefore reduces the angle that the force is applied along as compared to the axis 367 (e.g., as compared to the force being applied directly at the second end 345). A smaller angle means that the more of the force is directed along the horizontal axis 367 into the bias of the spring and less is directed toward the wall of the bore 315 (e.g., limiting frictional engagement).

(93) The pivoting movement of each push pin 370 enables the wobble plate 230 and inclined plate 220 to continue to move so that the system can maintain reciprocating motion. The push pins 370 are oriented so that in the driving position, the push pin 370 can transfer force closer to the first end 340 and achieve more efficiency, while in the non-driving position, the push pin 370 does not interfere with the motion of the system.

(94) Although the above description was specifically focused on a hydraulic tool, this disclosure could be used in similar reciprocating devices. For example, other types of systems that are driving by the movement of a piston within a cylinder may experience similar problems of inefficiency and frictional wear. A push pin inserted into a piston that includes an internal cavity can be readily used by a person of ordinary skill in the art in order to achieve a greater percentage of the resultant force directed along the direction of travel of the piston.

(95) One of ordinary skill will appreciate that the exact dimensions and materials are not critical to the disclosure and all suitable variations should be deemed to be within the scope of the disclosure if deemed suitable for carrying out the objects of the disclosure.

(96) One of ordinary skill in the art will also readily appreciate that it is well within the ability of the ordinarily skilled artisan to modify one or more of the constituent parts for carrying out the various examples of the disclosure. Once armed with the present specification, routine experimentation is all that is needed to determine adjustments and modifications that will carry out the present disclosure.

(97) The above examples are for illustrative purposes and are not intended to limit the scope of the disclosure or the adaptation of the features described herein. Those skilled in the art will also appreciate that various adaptations and modifications of the above-described preferred examples can be configured without departing from the scope and spirit of the disclosure. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described.

Claims

1. A reciprocating assembly for driving a working portion of a tool, the reciprocating assembly comprising: a bore configured to contain hydraulic fluid; a piston received within the bore and configured to move within the bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the piston within the bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the bore, the piston including an internal cavity extending at least partially along a length of the piston; and a push pin received within the internal cavity of the piston; wherein the bore, the piston, and the push pin are configured such that a movement of the push pin relative to the bore causes translation of the piston within the bore to the extended position along a translation axis; and wherein the push pin is configured to pivot during the movement of the push pin, between an in-line position and an angled position, the in-line position substantially parallel to the translation axis; and wherein the push pin includes a cutout at each of two opposite ends, and wherein each cutout is configured to at least partially receive a ball bearing, and wherein one of the ball bearings is received within the internal cavity and the other of the ball bearings is coupled to a drive assembly configured to supply a force which causes the movement of the push pin, and wherein the push pin is configured to rotate about the

ball bearings.

2. The reciprocating assembly of claim 1, wherein the cavity extends along a majority of the length of the piston.
3. The reciprocating assembly of claim 1, wherein the internal cavity includes a terminal end and an open end, the open end being wider than the terminal end.
4. The reciprocating assembly of claim 1, further comprising a drive assembly configured to supply a force which causes the movement of the push pin, wherein the drive assembly comprises: an electric motor; a gear assembly configured to be driven by the electric motor; an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly; and a wobble plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; wherein the wobble plate is configured to not rotate with the inclined plate.
5. The reciprocating assembly of claim 4, further comprising an anti-rotational assembly connected between the wobble plate and a housing containing the bore, the anti-rotational assembly configured to limit rotation of the wobble plate and permit axial movement of the wobble plate.
6. A reciprocating assembly for driving a working portion of a tool, the reciprocating assembly comprising: a bore configured to contain hydraulic fluid; a piston received within the bore and configured to move within the bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the piston within the bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the bore, the piston including an internal cavity extending at least partially along a length of the piston; and a push pin having a first end received within the internal cavity of the piston, the first end having a first radius of curvature; wherein the push pin is configured to pivot about a center of the first radius of curvature between an in-line position and an angled position, the in-line position substantially parallel to a translation axis of the piston; and wherein the piston is configured to move from the retracted position toward the extended position when the push pin is in the in-line position; wherein the first end includes a first cutout having the first radius of curvature; wherein the first cutout at least partially receiving a first ball bearing; and wherein the push pin is configured to rotate about the first ball bearing.
7. The reciprocating assembly of claim 6, wherein: a second end of the push pin opposite the first end includes a second cutout having a second radius of curvature; the second cutout at least partially receiving a second ball bearing; and the second ball bearing is coupled to a drive assembly configured to supply a force which causes the movement of the push pin.
8. The reciprocating assembly of claim 4, further comprising a drive assembly configured to supply a force which causes the movement of the push pin, wherein the drive assembly comprises: an electric motor; a gear assembly configured to be driven by the electric motor; an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly; and a wobble plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; wherein the wobble plate is configured to not rotate with the inclined plate.
9. The reciprocating assembly of claim 6, wherein an are angle of the first end is less than 180 degrees.
10. A reciprocating assembly for driving a working portion of a tool, the reciprocating assembly comprising: a bore configured to contain hydraulic fluid; a piston received within the bore and configured to move within the bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the piston within the bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the bore; and a push pin contacting the piston; a drive assembly configured to supply a force which causes the movement of the push pin, the drive assembly including a non-rotational plate configured to supply the force by rocking between a first plate position and a second plate position, the plate being closer to the piston in the first plate position than in the second position; wherein the bore, the piston, and the push pin are configured such that a movement of the push pin relative to the bore causes translation of the piston within the bore to the extended position along a translation axis; wherein the push pin is configured

to pivot during the movement of the push pin, between an in-line position and an angled position, the in-line position substantially parallel to the translation axis, the push pin being in the in-lined position when the plate is in the first plate position and the push pin being in the angled position in the second plate position; and wherein the push pin further includes a first end having a first concave surface and a second end having a second concave surface, wherein a first ball bearing is received within the first concave surface and contacts the piston, wherein a second ball bearing is received within the second concave surface and contacts the plate, and wherein the push pin is configured to rotate about the first ball bearing.

11. The reciprocating assembly of claim 10, wherein: the plate includes a third concave surface; and the second ball bearing is at least partially received within the third concave surface; wherein the second ball bearing configured to permit relative movement between the push pin and the plate.

12. The reciprocating assembly of claim 10, wherein the bore is a first bore, the push pin is a first push pin and a piston is a first piston, the reciprocating assembly further comprising: a second bore spaced apart from the first bore and configured to contain hydraulic fluid; a second piston received within the second bore and configured to move within the second bore in a reciprocating motion between an extended position and a retracted position, wherein displacement of the second piston within the second bore is configured to cause displacement of hydraulic fluid when hydraulic fluid is in the second bore; and a second push pin contacting the second piston; the drive assembly configured to supply a force which causes the movement of the second push pin.

13. The reciprocating assembly of claim 12, wherein the plate is closer to the first piston in the first plate position and closer to the second piston in the second position.

14. The reciprocating assembly of claim 12, wherein the translational axis is a first translational axis, wherein: the second piston moves along a second translational axis that is parallel to the first translational axis; and the second push pin is configured to pivot during the movement of the second push pin, between a second in-line position and a second angled position, the second in-line position substantially parallel to the second translation axis, the second push pin being in the second in-lined position when the plate is in the second plate position and the second push pin being in the angled position in the first plate position.

15. The reciprocating assembly of claim 1, further comprising an O-ring connected to an outer surface of the push pin, wherein the O-ring is compressible.

16. The reciprocating assembly of claim 6, further comprising an O-ring connected to an outer surface of the push pin, wherein the O-ring is compressible.

17. The reciprocating assembly of claim 6, wherein the push pin includes a hyperboloid shape.

18. The reciprocating assembly of claim 10, further comprising an O-ring connected to an outer surface of the push pin, wherein the O-ring is compressible.

19. The reciprocating assembly of claim 10, wherein the push pin includes a hyperboloid shape.

20. The reciprocating assembly of claim 10, wherein the drive assembly further includes: an electric motor; a gear assembly configured to be driven by the electric motor; an inclined plate rotatably connected to the gear assembly and configured to be driven by the gear assembly; and the non-rotational plate connected to the inclined plate and configured to be driven by rotation of the inclined plate; wherein the non-rotational plate is configured to not rotate with the inclined plate.
